

INSTITUTO SUPERIOR TÉCNICO

Master's in Engineering and Data Science

Predicting Atrial Fibrillation Following Myocardial Infarction: A Comparison of Naive Bayes and k-NN

Statistical Methods in Data Mining

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1 Introduction

The aim of this project is to predict whether a patient develops atrial fibrillation (AF) as a complication after a myocardial infarction, using only the information available at hospital admission (Question 1), and to implement two different classification methods, naive Bayes and k-nearest neighbours (k-NN), for this task.

The data is the *Myocardial infarction complications* dataset from the UCI Machine Learning Repository: 1700 patients and 124 columns (an ID, 111 inputs recorded at admission and during the first three days and 12 complications). Missing values are coded as `?`. Question 1 concerns the information available at hospital admission, so we use inputs 2-112 except the pain relapses and drugs given during the first days (columns 93-95 and 100-105). This leaves 102 inputs: 78 binary, 12 ordinal (coded levels) and 12 continuous (age, blood pressure, laboratory values). The response variable is atrial fibrillation as a complication (`FIBR_PREDS`, column 113). The other complications are not used as input.

2 Exploratory data analysis

2.1 First data audit

A first data audit was performed on the whole data set. The audit only checks whether values are valid, it does not use the response and estimates nothing from the data, so it causes no leakage. It found no duplicated patients, no binary input with values other than 0/1 and ordinal codes within the documented ranges. However, 34 blood-pressure values were set to 0 mmHg, which is physiologically impossible. These values should therefore be treated as missing. This type of cleaning will also have to be done to future patients, and was therefore applied before the split.

2.2 Class balance and train/test split

The data was then split into a training set (n = 1360) and a test set (n = 340), with an 80/20 ratio, chosen to maximise the amount of data available for fitting and validation, at the cost of a potentially higher variance of the test metric. This split was stratified by AF (response variable) so both contain 10% AF cases. This class imbalance makes accuracy misleading. A classifier that always predicts "no AF" would reach 90% accuracy without detecting a single AF case. Preprocessing configurations and neighbourhood sizes were selected using cross-validated ROC-AUC. Decision thresholds were selected by maximizing Youden's index, equivalently balanced accuracy. Evaluation included ROC-AUC, accuracy, zero–one error, sensitivity, specificity, balanced accuracy, precision and F1-score. All further exploratory analysis and decisions were based on the training set only, and the test set was reserved for the final evaluation of the classifiers. The variables of the training set were grouped into three separate categories: *binary variables, ordinal variables*, and *continuous variables*.

2.3 Missing data

Of the 102 inputs, 101 have at least one missing value, and every training patient lacks at least one input, so complete-case analysis using all inputs is impossible (Figure A.1). Four inputs have more than 50% missing values: serum creatine phosphokinase (`KFK_BLOOD`, 99.7%), family history of coronary heart disease (`IBS_NASL`, 95.9%), and the two blood pressure measurements by the emergency team (`S_AD_KBRIG`: 63.7%; `D_AD_KBRIG`: 64.0%). The next most incomplete inputs are three drugs given by the emergency team (`NOT_NA_KB`, `LID_KB`, and `NA_KB`, approximately 39–40%), so no input has between 40% and 63% missing values.

Among the remaining continuous inputs, age is almost complete (0.4% missing), while serum potassium and sodium (`K_BLOOD`, `NA_BLOOD`) are missing most often (21.3% and 21.5%). ICU blood pressure (`S_AD_ORIT`, `D_AD_ORIT`) is missing for 15.5% and 16.1% of patients, respectively.

Missingness also shows a block structure. The 26 inputs describing the admission ECG (rhythm, rhythm disturbances and conduction disturbances) tend to be missing together. For 90 training patients, no information was recorded for any of these ECG inputs. Furthermore, 71 patients lack more than 30% of all inputs; 66 of these patients have no recorded ECG information, and six developed AF.

Missingness may also be related to the response. Patients whose ICU diastolic blood pressure was missing developed AF less often than those with a recorded value (4.1% versus 11.1% for `D_AD_ORIT`; Fisher's exact test, $p = 0.0008$). The association remained significant at the 5% level after correction

for the 49 tests (FDR-adjusted $q = 0.04$). This provides evidence against the assumption that the data are missing completely at random. Whether missingness is MAR or MNAR cannot be determined from these observed associations alone.

2.4 Binary variables

Most binary inputs are rarely present. The median share of patients with value 1 is 1.9% for ECG findings at admission and 1.0% for medical history indicators, while drugs are more common (median 38%). Thirty binary inputs are present in fewer than about 1% of patients. None showed an association with AF at $p < 0.05$, although their rarity limits the ability to detect such associations. Patients with at least one of these indicators had a similar AF rate to the remaining patients (10.3% versus 10.0%). Estimated associations can also be unstable: an indicator present in a single patient who developed AF produces an infinite uncorrected odds ratio. Sparse category counts can lead to zero estimated class-conditional probabilities in Naive Bayes, which were addressed using Laplace smoothing.

Each binary input was tested for association with AF using Pearson's chi-squared test when all expected counts (the counts expected if the input were unrelated to AF) are at least 5, otherwise Fisher's exact test was used, since the chi-squared approximation is unreliable for small counts. Of the 76 tested inputs (the two remaining binary inputs are almost entirely missing or constant), nine are associated with AF at p < 0.05 (See Table A.1). Because 76 tests are run, a few could be chance findings and after a Benjamini-Hochberg false discovery rate (FDR) correction, eight remain (q < 0.05) significant. The exception is normal sinus rhythm on the admission ECG (`ritm_ecg_p_01`, p = 0.044, q = 0.37), whose association is weak. Six of the nine concern AF or arrhythmia already present before or at admission. In addition, a history of bronchial asthma (`zab_leg_03`) is associated with more AF and male sex (`SEX`) with less.

Six inputs describing AF before or at admission was put under a closer look: AF paroxysms (`nr_03`), persistent AF in the history (`nr_04`), AF paroxysms at ICU admission (`MP_TP_POST`), AF rhythm on the admission ECG (`ritm_ecg_p_02`) and AF paroxysms (`n_r_ecg_p_05`) and persistent AF (`n_r_ecg_p06`) on the admission ECG. Patients with at least one of them (n = 123) had an AF rate of 32.5% against 7.8% for the rest, but they account for only 29% of the AF cases. Persistent AF (`nr_04` and `n_r_ecg_p_06`) was associated with a lower AF rate (1 of 23 and 1 of 24 patients), possibly because ongoing AF is not recorded as a new complication.

2.5 Ordinal variables

Twelve inputs are ordinal. Missingness is low ($0.1 - 15\%$). Four levels contain fewer than 17 training patients, angina functional class IV (`FK_STENOK` = 4, n = 8), hypertension stage 1 (`GB` = 1, n = 10), heart-failure level 4 (`ZSN_A` = 4, n = 15) and a QS pattern in the posterior wall (`post_im` = 4, n = 10). Their AF rates are very uncertain as there are few patients . All ordinal inputs are only weakly related to AF (|Spearman ρ| between 0.002 and 0.07) (Table A.3).

To decide how to code the ordinal inputs, we examined how the AF rate changes across their levels. Ordinal levels are ordered, but the differences between them are not necessarily equal. Using their original numerical codes in distance calculations would impose equal spacing between consecutive codes. The data do not support this assumption. All ordinal inputs have a weak rank correlation with AF (|Spearman $\rho| \leq 0.07$) and for most of them the AF rate varies across levels without a steady direction. Except for `IBS_POST`, there is no evidence of any trend, so categorical coding costs little and avoids assuming equal spacing (Appendix, figure A.3). The clearest case is coronary heart disease in the weeks before admission (`IBS_POST`), with AF rates of 10.6%, 13.6% and 6.7%. The levels are different types of angina rather than steps on a scale. The AF rate differs clearly between the levels (chi-squared test, $p = 0.001$), although the rank correlation is weak ($\rho = -0.07$). A numeric coding, would largely miss this pattern.

2.6 Continuous variables

Twelve of the 102 inputs are continuous. For this EDA, three of them were removed for missingness: serum CPK (`KFK_BLOOD`) and the systolic and diastolic blood pressure measured by the emergency team (`S_AD_KBRIG`, `D_AD_KBRIG`). This section describes the remaining nine on the training set: age (`AGE`), systolic and diastolic blood pressure in the ICU (`S_AD_ORIT` and `D_AD_ORIT`), serum potassium (`K_BLOOD`) and sodium (`NA_BLOOD`), the liver enzymes ALT and AST (`ALT_BLOOD`, `AST_BLOOD`), the white blood cell count (`L_BLOOD`) and the erythrocyte sedimentation rate (`ROE`).

The distribution of each input is shown in Figure A.4, with two histogram bin widths and a boxplot with the individual observations. `AGE` and `NA_BLOOD` are roughly symmetric (skewness -0.24 and -0.16) and `K_BLOOD` is mildly right-skewed (skewness 0.91). `ALT_BLOOD`, `AST_BLOOD`, `ROE`, and `L_BLOOD` are clearly right-skewed. A log transform makes `L_BLOOD` nearly symmetric, while for `ALT_BLOOD`, `AST_BLOOD`, and `ROE` it reduces the skew but reveals gaps between values, because the values are recorded with few decimal places (Figure A.5). Blood pressure (`S_AD_ORIT` and `D_AD_ORIT`) is recorded in steps of 10 $mmHg$, which produces the gaps in the 40-bin histograms. In addition to this, `D_AD_ORIT` is strongly concentrated at 80 $mmHg$ (its first quartile equals its median).

For continuous variables, the association with AF was also examined. Figure A.6 and Table A.4 compare each input between patients with and without AF, and Figure A.7 shows the point-biserial correlation of each input with AF. Figure A.8 shows the Empirical Cumulative Distribution Functions for each continuous variable with and without AF. `AGE` shows the clearest difference, with a median of 66 with AF and of 62 without. The analysis shows there are very few patients with AF under 50 years old. The correlation of age with AF (0.14) is the largest for the continuous variables. `K_BLOOD` is the second most associated input (-0.07), having slightly lower and less dispersed values in AF patients. `ROE` and `L_BLOOD` have correlations of about 0.03. Each of these comparisons uses only the patients with that input recorded, and overall the continuous inputs separate the two classes only weakly.

2.7 Correlation between inputs

Correlations between inputs were computed with Spearman's rank correlation (ρ) on the observed values. We chose Spearman since it uses only the order of the values, so it does not treat ordinal codes as equally spaced steps. It is also robust to skewness and outliers.

For two binary inputs it gives exactly the same value as Pearson's correlation, the phi coefficient, which summarises their contingency table, so nothing changes for the binary inputs.

Table A.2 shows the association between six pairs of ordinal or binary inputs with $|\rho| \geq 0.7$. This result is expected as the description and medical conditions correlate closely. For example, the anterior and inferior infarct patterns which tend to exclude each other. Such pairs matter for classification because dependence within each class can violate the conditional independence assumption of Naive Bayes, while related predictors can give repeated weight to similar information in the Gower distance of k-NN. This is particularly relevant for AF paroxysms at ICU admission (`MP_TP_POST`) and AF rhythm on the admission ECG (`ritm_ecg_p_02`), $\rho = 0.85$, and normal sinus rhythm (`ritm_ecg_p_01`) vs. sinus tachycardia (`ritm_ecg_p_07`), $\rho = -0.78$. The latter are mutually exclusive categories of an ECG variable: the two indicators cannot both equal 1, although they can both equal 0.

The strongest correlations between ordinal inputs (Spearman, observed values) are angina functional class (`FK_STENOK`), $\rho = 0.85$, and hypertension stage (`GB`) vs. its duration (`DLIT_AG`), $\rho = 0.79$ (Appendix, Figure A.2).

Figure A.9 shows the correlation between the continuous variables. The only strong association for continuous variables is between `S_AD_ORIT` and `D_AD_ORIT`, with a correlation of 0.84.

2.8 Implications for modelling

Since every patient has at least one missing input, missing values will need to be handled, for example by imputation. Because the ECG inputs tend to be missing together and missingness of the ICU blood pressure is related to AF, the missingness pattern itself may carry information (see subsection 3.4). Several laboratory values (`ALT_BLOOD`, `AST_BLOOD`, `ROE` and `L_BLOOD`) are clearly right-skewed, which matters for both classifiers. Gaussian naive Bayes assumes normally distributed inputs within each class, and the range scaling used in the Gower distance of k-NN is sensitive to long right tails, so a log transformation could be considered. Most binary inputs are rare, and rare inputs and sparse ordinal levels can give zero counts in a class, which should be kept in mind for naive Bayes (for instance, through Laplace smoothing). As the AF rate does not change steadily across the levels of the ordinal inputs, treating them as categorical rather than numerical seems more appropriate. The inputs describing AF before or at admission are the strongest predictors among the binary inputs, but part of their effect may reflect AF that is already present rather than new AF, which is worth considering when interpreting the results. Finally, strongly correlated pairs, in particular `MP_TP_POST` and `ritm_ecg_p_02`, may challenge the Naive Bayes conditional independence assumption if dependence persists within each class and would get double weight in the Gower distance of k-NN, so their joint use should be considered carefully.

3 Preprocessing

The training set (subsection 2.2) was used for 5-fold cross-validation, with a stratified rotating split of 4/5 for fitting and 1/5 for validation. All preprocessing steps that learn from the data (filters, imputation, category levels) are fitted only on the fitting data of each fold and then applied, unchanged, to the fitting and validation data. In the final model, they are fitted on the full training set and applied to the test set. Thus, validation and test observations were not used to estimate fold-specific preprocessing parameters. The steps, in order of application, are summarised in Table 1.

Table 1: Preprocessing steps, in order of application.

#	Step	Learned from	Applied to	Model
1	Blood pressure = 0 mmHg → missing value	fixed rule	original data, before the split	both
2	Stratified 80/20 split; 5-fold CV on the training set	–	–	–
3	Logarithm of `ALT_BLOOD`, `AST_BLOOD`, `L_BLOOD` and `ROE`	choice of variables in the EDA (training)	training and test	both
4	Filters 1–8 (Table 2)	fitting data	fitting, validation, test (filter 2: fitting only)	both (with their own thresholds)
5	Imputation (median, mode)	retained fitting data	fitting, validation, test	both
6	Missing-value indicators (ICU BP, admission ECG), computed before imputation	kept if not constant in the fitting data	fitting, validation, test	NB only
7	Scaling $(x - \min)/\text{range}$ (differences capped at 1 in the Gower distance)	fitting data	fitting, validation, test	k-NN only
8	Outlier identification (robust Mahalanobis)	fitting data	diagnostic only; nothing removed	–

3.1 Fixed preprocessing

Before splitting the original data, blood pressure values equal to 0 mmHg, which are physiologically impossible, were converted to missing values.

Natural-log transformations were applied to `ALT_BLOOD`, `AST_BLOOD`, `L_BLOOD` and `ROE`, which were chosen in the EDA using the training data only: they are the positive continuous variables with skewness greater than 1. The transformation is applied sample by sample, in the same way to the training and test sets, and is prevented from being applied twice.

3.2 Variable and sample selection filters

A sequence of eight filters was applied (Table 2). Each filter starts from the variables and samples kept by the previous ones. Variables with no association with the response variable produce spurious associations (noise). Since the k-NN classifier depends on the geometric integrity of the distances and Naive Bayes on the computation of probabilities, this noise wrongly influences model fitting. Redundant variables repeat the same information, distorting the geometric distances by giving excessive weight to duplicated axes. Constant variables add no information and increase the computational cost.

Table 2: Filters applied, in order, and values tested for each threshold (t). **Filter 2** thresholds were chosen from EDA on the training data (80%), which found that only one sample had more than 50% missing values. The thresholds of **filters 5 and 6** were chosen to test the same levels of evidence. Formulas used are listed in Appendix B.

#	Filter	Applies to	Removal rule	Values tested
1	Missing values (columns)	all variables	variables with proportion of missing values $> t$	OFF, 0.25, 0.50, 0.75
2	Missing values (rows)	fitting samples	samples with proportion of missing values $> t$	OFF, 0.25
3	Constants	all variables	variables with a single observed value	always ON
4	Dispersion	continuous	MAD / range $\leq t$, with $b = 1.4826$	OFF, 0.005, 0.01, 0.03
5	Feature–target correlation	continuous	Pearson $\lvert r \rvert$ with the target $\leq t$	OFF, 0.01, 0.02, 0.05
6	Feature–target association	categorical	p-value of the χ^2, Fisher or Monte Carlo χ^2 test $\geq t$	OFF, 0.75, 0.50, 0.10
7	Feature–feature correlation	continuous	in each pair with $\lvert r \rvert > t$, the variable less correlated with the target (pair `S_AD_ORIT` / `D_AD_ORIT` not evaluated)	OFF, 0.80, 0.85, 0.90
8	Feature–feature association	categorical	in each pair with bias-corrected Cramér's $V > t$, the variable less associated with the target	OFF, 0.80, 0.85, 0.90

3.3 Optimisation of the filters thresholds

The thresholds were optimised independently for each classifier by grid search ($4^6 \times 2 = 8192$ configurations), evaluating each configuration on the 5 folds. For k-NN, the configuration and the number of neighbours k were chosen jointly. Except for filter 3, all filters can be switched off (OFF), i.e. the optimisation can choose to exclude the filter. The configuration with the highest AUC (Area Under the ROC Curve), computed from the pooled out-of-fold predictions of the 1360 samples, is selected. In case of a tie, the first configuration in grid order is selected. Finally, the selected configuration is applied once to the whole training set (1360 samples). As a reference, the original configuration, with thresholds fixed in advance (baseline), was also evaluated. Table 3 shows the chosen configuration and its effect on the final fits.

Table 3: Selected thresholds and effect on the final fits (full training set).

	Baseline (both)	Optimised NB	Optimised k-NN
Filter 1 (missing values - columns)	0.50	0.50	0.75
Filter 2 (missing values - rows)	0.50	0.25	0.25
Filter 4 (dispersion)	0.01	OFF	OFF
Filter 5 (feature-target correlation)	0.02	0.05	0.05
Filter 6 (feature-target association)	0.50	0.10	0.10
Filter 7 (feature-feature correlation)	0.85	OFF	0.80
Filter 8 (feature-feature association)	0.85	OFF	OFF
k (k-NN)	121	–	25
Nº Variables removed by filters (1 / 3 / 5 / 6)	4 / 1 / 5 / 43	4 / 1 / 7 / 71	2 / 1 / 9 / 71
Original variables kept	49	19	19
Training patients removed (filter 2)	1	97 (13 with AF)	96 (13 with AF)

Note: filters 4, 7 and 8 did not remove any variable in any of the models.

The two classifiers converged to almost identical configurations: they chose the same values for five of the seven filters and differ only in filters 1 and 7. Both optimized configurations retained the same 19 original variables, but their retained fitting patients and predictions differed. Compared with the baseline, the optimized configuration is stricter on relevance (filters 5 and 6 at their strictest level) and removes from the fitting data the samples with more than 25% missing values. Filter 6 is the one that reduces dimensionality the most: it removes 43 variables in the baseline and 71 in the optimised versions. The sensitivity of the AUC to each filter is shown in Figure B.1.

For k-NN, the selected number of neighbours dropped from $k = 121$ (baseline, 49 variables) to $k = 25$

(optimised, 19 variables). With the other thresholds fixed at their selected values, the best k was 61 with filter 6 disabled and 25 with its threshold set to 0.10. This comparison does not isolate the contribution of filter 6 to the overall baseline–optimized difference (Figure B.1).

Table 4: Cross-validation AUC (pooled out-of-fold) without and with filters. Filter 3 (constant predictors) remained active in all configurations.

	Original variables	AUC NB	AUC k-NN (k)
No filters (Optional filters OFF)	101	0.668	0.638 (81)
Baseline	49	0.695	0.629 (121)
Optimised	19	0.713	0.688 (25)

In validation, the optimised filters improved the AUC of both models compared with using no filters (from 0.668 to 0.713 for naive Bayes and from 0.638 to 0.688 for k-NN). The baseline filters, in contrast, improved naive Bayes (0.695) but slightly reduced the AUC of k-NN (0.629). Since the optimised configuration is the best of 8192 candidates, its validation AUC is optimistic. Performance on new patients is assessed on the test set (results section).

3.4 Missing values and preparation for the models

After the filters, the remaining inputs still contain missing values, so they must be handled before the classifiers can be fitted. The EDA showed an association between missing ICU blood pressure records and a lower AF rate. Thus, the data is not missing completely at random (subsection 2.3). Whether the data is missing at random (MAR) or not at random (MNAR) cannot be determined from the observed data, so we assume MAR.

Therefore, in Naive Bayes, indicators for missing ICU blood pressure and missing admission ECG records were considered, in order to preserve information about the missing-value pattern. A single indicator is used for the ECG block, since 26 identical indicators would be counted 26 times by naive Bayes. Only the indicators that were not constant in the fitting data were kept. In k-NN, these indicators were not included; scaling by the range of the fitting data prevents the units of measurement from determining the weight of the numerical variables in the Gower distance.

The imputation is performed after the filters only on the remaining variables. The median (for continuous variables) and the mode (for ordinal and binary variables) were computed only on the fitting data of each fold and applied to the fitting and validation data of that fold. In the final model, they are recomputed on the full training set and applied to the test set.

Among the 49 inputs tested, no other missingness association remained significant after FDR correction (all other $q \geq 0.40$, Appendix Table B.1). This does not rule out bias introduced by imputation.

3.5 Outlier detection

Extreme observations may reflect the disease itself rather than recording errors. Since the two cannot be reliably distinguished without clinical reference ranges for each variable, outliers were neither removed nor modified, so as not to remove real biological variability. Outliers were only identified, with the baseline configuration, in each fold, using the robust Mahalanobis distance, with location and covariance estimated by the minimum covariance determinant (MCD, $\alpha = 0.75$):

$$\mathrm{RD}_i^2 = (x_i - \hat{\mu}_{\mathrm{MCD}})^\top \hat{\Sigma}_{\mathrm{MCD}}^{-1} (x_i - \hat{\mu}_{\mathrm{MCD}}) > \chi^2_{p,\,0.975}. \tag{1}$$

The threshold used was the 97.5% quantile of the chi-squared distribution with p degrees of freedom, where p is the number of continuous variables retained in the fitting data of each fold (between 5 and 7). This threshold is an approximate reference under multivariate normality. Between 6.8% and 10.5% of the fitting patients were flagged per fold. These proportions exceed the nominal reference of 2.5%, but do not, by themselves, constitute a formal test of normality (Table B.2).

4 Classification methods

4.1 Method 1: Naive Bayes

4.1.1 Method overview

Naive Bayes is a probabilistic classifier that assumes conditional independence between predictors given the outcome [2]. For a patient with predictor vector $\mathbf{x} = (x_1, \ldots, x_p)$,

$$\widehat{P}(Y = c \mid \mathbf{x}) = \frac{\widehat{\pi}_c \prod_{j=1}^{p} \widehat{f}_{jc}(x_j)}{\sum_{h \in \{0,1\}} \widehat{\pi}_h \prod_{j=1}^{p} \widehat{f}_{jh}(x_j)},$$

where $Y = 1$ denotes atrial fibrillation, $\widehat{\pi}_c$ is the estimated class prior, and $\widehat{f}_{jc}$ is a class-conditional density or probability mass function.

The positive-class score is $s(\mathbf{x}) = \widehat{P}(Y = 1 \mid \mathbf{x})$. AF is predicted when $s(\mathbf{x}) \geq t$. The method offers simple estimation, low computational cost and support for numerical and categorical predictors. Its probability estimates nevertheless depend on the independence and distributional assumptions and were not assessed for calibration.

4.1.2 Predictor distributions

The implementation used `e1071::naiveBayes` with `laplace = 1` [4]. Numerical predictors were modelled using class-specific Gaussian distributions:

$$X_j \mid Y = c \sim \mathcal{N}(\mu_{jc}, \sigma_{jc}^2).$$

Means and variances were estimated from the retained fitting observations, after the shared transformations and imputation.

Binary and ordinal-coded predictors were converted into categorical factors. Ordinal ordering was therefore not used. Class-conditional category probabilities were estimated with Laplace smoothing, preventing zero probabilities for fitting-set levels absent from a particular class. The smoothing formula and treatment of unseen categories are provided in Appendix C.1.

Class priors were estimated from the retained fitting patients. No class reweighting or resampling was applied.

4.1.3 Training and model selection

Model parameters were estimated independently within each fitting partition. The selected configuration, ID 231, achieved validation AUC 0.7133, compared with 0.6953 for the baseline. Exact AUC ties retained the first configuration in grid order. The baseline was evaluated separately: its filter-2 threshold of 0.50 was outside the searched values.

After selection, both configurations were refitted using the training partition. The baseline retained 1359 patients and 51 predictors: 49 original variables and two missingness indicators. The optimized configuration retained 1263 patients and 20 predictors: 19 original variables and the ICU blood pressure missingness indicator. Its ECG missingness indicator was constant among retained fitting patients and was excluded.

4.1.4 Decision threshold and evaluation

Both the conventional threshold $t = 0.5$ and validation-selected Youden thresholds were evaluated. The latter maximized

$$J(t) = \text{Sensitivity}(t) + \text{Specificity}(t) - 1,$$

which is equivalent to maximizing balanced accuracy, rather than minimizing ordinary zero–one error [5, 6]. Among tied finite thresholds, the largest was retained.

The selected thresholds were 0.05334437 for the baseline and 0.04069238 for the optimized configuration. The code used unrounded values, applied unchanged to the test set.

Evaluation included ROC-AUC, accuracy, zero–one error, sensitivity, specificity, balanced accuracy, positive-class precision and F1-score. Test AUC uncertainty was summarized using 95% DeLong confidence intervals [5]. Because configuration and threshold selection used the same validation predictions, development performance may be optimistic. The procedure was not nested cross-validation.

4.2 Method 2: k-Nearest Neighbours

4.2.1 Method overview

The k-nearest neighbours (k-NN) classifier predicts a patient's outcome using the outcomes of similar fitting patients [2]. For predictor vector $\mathbf{x}$, the k patients with the smallest dissimilarities form the neighbourhood $\mathcal{N}_k(\mathbf{x})$. The positive-class score is

$$s_k(\mathbf{x}) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\mathbf{x})} \mathbb{I}(y_i = 1),$$

where $y_i = 1$ denotes atrial fibrillation. Neighbours receive equal voting weight, and AF is predicted when $s_k(\mathbf{x}) \geq t$. Distance ties are resolved by fitting-row order, ensuring exactly k neighbours.

The method can represent nonlinear decision boundaries without specifying a parametric relationship between predictors and outcome. It requires neither Gaussian predictors nor conditional independence. However, it relies on the chosen representation producing informative neighbourhoods. Its scores describe local positive-class frequencies and were not assessed for calibration.

4.2.2 Dissimilarity measure and predictor representation

A Gower-style dissimilarity combined numerical and categorical predictors [3]. For patients $\mathbf{x}$ and $\mathbf{z}$,

$$d(\mathbf{x}, \mathbf{z}) = \frac{1}{p} \sum_{j=1}^{p} \delta_j(x_j, z_j),$$

where p is the number of retained predictors. Each predictor receives equal weight.

For numerical predictors,

$$\delta_j(x_j, z_j) = \min \left\{ \frac{|x_j - z_j|}{R_j}, 1 \right\},$$

where R_j is the range among retained fitting patients, after transformation and imputation. Validation and test values may fall outside this range; the cap applies to distance contributions, rather than to the observations themselves.

For categorical predictors, $\delta_j(x_j, z_j) = \mathbb{I}(x_j \neq z_j)$. Ordinal codes were treated as nominal, discarding their ordering. Binary predictors were symmetric: shared zeros and shared ones both contributed zero dissimilarity. Generated missingness indicators were not included in k-NN.

Missing values were imputed using fitting-set medians or modes. Filtering, imputation and numerical ranges were estimated within each fitting partition and applied unchanged to validation observations. Range scaling remains sensitive to extreme fitting values; capping contributions does not make it robust.

4.2.3 Cross-validation and parameter selection

Using the shared split and five-fold validation procedure, the baseline was evaluated separately from a search over 8192 preprocessing configurations. For the baseline and each searched configuration, candidate neighbourhood sizes were

$$k \in \{3, 5, 9, 15, 25, 41, 61, 81, 121, 181, 251, 351\}.$$

Candidates were compared using ROC-AUC calculated from pooled out-of-fold scores. Preprocessing used only the fitting partition, and no validation patient was removed by the fitting-row missingness filter.

The baseline selected $k = 121$, with validation AUC 0.6291. Configuration 744 selected $k = 25$, with validation AUC 0.6884. Its filter thresholds are reported in the shared preprocessing section. The baseline was outside the search grid because its filter-2 threshold of 0.50 was not among the searched values. The effect of k under both preprocessing configurations is shown in Appendix D.1.

Exact AUC ties retained the first configuration in grid order. Within a configuration, the smallest tied k was retained because candidates were evaluated in ascending order. No random tie-breaking was used.

Youden thresholds were selected from the corresponding pooled out-of-fold scores, using the criterion described for Naive Bayes. The selected values were 0.08677686 for the baseline and 0.10 for the optimized

configuration, requiring at least 11 positive neighbours out of 121 and three out of 25, respectively. The code used unrounded values.

The final baseline reference set contained 1359 patients and 49 predictors; the optimized version contained 1264 patients and 19 predictors. Their preprocessing settings, k values and thresholds were fixed before test evaluation. Selection used the same validation folds throughout, so development performance may be optimistic.

5 Results and discussion

5.1 Naive Bayes

The baseline and optimized configurations were evaluated on the same reserved test set of 340 patients, including 34 positive AF cases. Preprocessing settings and validation-selected thresholds were held fixed throughout test evaluation.

5.1.1 Fitting and validation performance

Table 5 compares apparent fitting performance with validation performance. Fitting AUC is summarized by the mean and sample standard deviation across the five fitting partitions. Validation AUC is calculated from pooled out-of-fold predictions.

Table 5: Naive Bayes apparent fitting AUC (mean ± sample standard deviation across folds) and pooled out-of-fold validation AUC.

Configuration	Apparent fitting	Validation
Baseline	0.8160 ± 0.0087	0.6953
Optimized	0.7833 ± 0.0156	0.7133

Both configurations performed better on their fitting observations than on held-out observations. The optimized version achieved higher validation AUC and a smaller descriptive fitting–validation gap. However, these summaries use different aggregation procedures, and validation results also supported configuration selection. The smaller gap therefore does not establish superior generalization; the reserved test set provides the subsequent assessment.

5.1.2 Test-set performance

Table 6 reports test performance at the validation-selected Youden thresholds. AUC uncertainty was summarized using 95% DeLong confidence intervals calculated with `pROC` [5].

Table 6: Naive Bayes test-set performance at the validation-selected Youden thresholds.

Measure	Baseline	Optimized
Decision threshold	0.05334437	0.04069238
ROC-AUC	0.7598	0.7141
95% DeLong CI	[0.6924, 0.8272]	[0.6329, 0.7954]
Accuracy	0.6324	0.5265
Zero–one error	0.3676	0.4735
Sensitivity	0.7647	0.7647
Specificity	0.6176	0.5000
Balanced accuracy	0.6912	0.6324
Precision (AF class)	0.1818	0.1453
F1-score	0.2938	0.2441

The baseline achieved a higher test AUC than the optimized configuration, so the validation improvement did not translate into better test-set discrimination. The confidence intervals indicate substantial uncertainty, and a paired analysis would be required to establish a statistically significant difference.

The baseline also achieved higher specificity, balanced accuracy, precision and F1-score. However, precision remained low for both configurations: only 18.2% and 14.5% of positive predictions corresponded

to patients with AF. Although these values exceeded the 10% test-set prevalence, most positive predictions were false positives.

5.1.3 Confusion matrix analysis

Both configurations detected 26 of the 34 positive patients and missed eight, giving 76.5% sensitivity. Among the 306 negative patients, the baseline correctly classified 189 and produced 117 false positives. The optimized version correctly classified 153 and produced 153 false positives.

The baseline therefore generated 36 fewer false positives while achieving the same sensitivity. This explains its higher specificity and precision at these operating points. Complete confusion matrices for both thresholds are presented in Appendix C.3.

5.1.4 Effect of the decision threshold

At $t = 0.5$, the baseline detected seven positive patients and the optimized model detected five, giving sensitivities of 20.6% and 14.7%. Both produced 18 false positives and achieved 94.1% specificity. Their accuracies were 86.8% and 86.2%, despite missing most positive cases.

The lower Youden thresholds increased sensitivity to 76.5%, but false positives increased to 117 for the baseline and 153 for the optimized configuration. The improvement in positive-case detection therefore came at the expense of specificity and precision.

Youden's criterion maximizes balanced accuracy without incorporating application-specific misclassification costs. These thresholds should consequently be interpreted as validation-selected operating points rather than established clinical decision rules. Detailed results at $t = 0.5$ are provided in Appendix C.2.

5.2 k-Nearest Neighbours

The baseline ($k = 121$) and optimized ($k = 25$) configurations were evaluated on the same 340 reserved test patients. Preprocessing, neighbourhood sizes and validation-selected thresholds were fixed before test evaluation.

5.2.1 Fitting and validation performance

Table 7 compares fitting and validation AUC. Apparent fitting performance includes each queried patient's own observation as an eligible neighbour. The self-excluded diagnostic removes that reference observation from its neighbourhood.

Table 7: k-NN apparent fitting AUC (mean $\pm$ sample standard deviation across folds), mean self-excluded fitting AUC, and pooled out-of-fold validation AUC.

Configuration	Apparent fitting	Self excluded	Validation
Baseline	0.7166 ± 0.0073	0.6439	0.6291
Optimized	0.7948 ± 0.0113	0.6564	0.6884

Apparent fitting AUC exceeded validation AUC for both configurations. However, this gap cannot be attributed entirely to overfitting: a patient's own outcome can contribute directly to its apparent fitting prediction. Excluding that observation substantially reduced fitting AUC.

The self-excluded results are diagnostic, rather than complete leave-one-out validation, because preprocessing still used all retained fitting patients. Furthermore, fitting results are fold averages whereas validation AUC uses pooled scores, so their comparison is descriptive. The higher optimized validation AUC also reflects selection among many candidate configurations.

5.2.2 Test-set performance

Table 8 presents test performance at the validation-selected Youden thresholds. The 95% AUC confidence intervals were calculated using DeLong's method in `pROC` [5].

Table 8: k-NN test-set performance at the validation-selected Youden thresholds.

Measure	Baseline	Optimized
Decision threshold	0.08677686	0.10
ROC-AUC	0.7071	0.6762
95% DeLong CI	[0.6138, 0.8005]	[0.5755, 0.7769]
Accuracy	0.6118	0.6912
Zero–one error	0.3882	0.3088
Sensitivity	0.7353	0.6176
Specificity	0.5980	0.6993
Balanced accuracy	0.6667	0.6585
Precision (AF class)	0.1689	0.1858
F1-score	0.2747	0.2857

The validation improvement did not translate into higher test AUC: baseline discrimination was higher on this sample. A paired analysis would be required to establish a statistically significant difference.

The optimized configuration achieved higher accuracy, specificity, precision and F1-score, but lower sensitivity and slightly lower balanced accuracy. Precision remained limited for both versions, with most positive predictions corresponding to patients without AF.

5.2.3 Confusion matrix analysis

The baseline identified 25 of the 34 positive patients and missed nine, while correctly classifying 183 negative patients and producing 123 false positives. The optimized configuration identified 21 positive patients and missed 13, while correctly classifying 214 negatives and producing 92 false positives.

Optimization therefore resulted in 31 fewer false positives but four additional missed positive cases at the selected operating points. This illustrates the trade-off between sensitivity and specificity. Complete confusion matrices are provided in Appendix C.3.

5.2.4 Effect of the decision threshold

At $t = 0.5$, both configurations classified every test patient as negative, achieving 90% accuracy but zero sensitivity, 50% balanced accuracy and an F1-score of zero. Precision was undefined because there were no positive predictions.

The lower Youden thresholds enabled positive-case detection by accepting more false positives. This demonstrates why accuracy alone is insufficient for this imbalanced problem. Threshold selection did not incorporate clinical misclassification costs, so the resulting operating points require further assessment before any clinical application.

5.3 Comparison of the classification methods

Figure E.1 compares validation and test ROC curves for all four configurations. Both methods improved pooled validation AUC after preprocessing optimization: from 0.6953 to 0.7133 for Naive Bayes and from 0.6291 to 0.6884 for k-NN. Neither improvement translated into higher test AUC.

Baseline Naive Bayes achieved the highest observed test AUC (0.7598) and balanced accuracy (0.6912) at its validation-selected Youden threshold. It identified 26 of the 34 positive patients, compared with 25 for baseline k-NN and 21 for optimized k-NN. Optimized Naive Bayes also identified 26 positive patients, but produced more false positives than its baseline.

Optimized k-NN achieved the highest specificity at the Youden operating points (0.6993), accompanied by lower sensitivity (0.6176). No configuration was therefore best on every measure. The preferred operating point would depend on the relative consequences of false positives and false negatives.

The differences between validation and test results are consistent with selection optimism and sampling variability, although this analysis cannot distinguish their contributions. The baseline was also outside the preprocessing search grid, so the grid-selected configurations were not guaranteed to outperform it.

These comparisons are descriptive: statistical claims about AUC differences would require a paired analysis. Furthermore, the test set contained only 34 positive cases, so changing the prediction for one positive patient changes sensitivity by approximately 2.9 percentage points. Test results were used to assess the fixed configurations, without further tuning.

6 Conclusion

Using only information available at hospital admission, both classifiers predicted post-infarction atrial fibrillation better than chance, but only moderately well. Naive Bayes with the baseline preprocessing had the highest test ROC-AUC (0.76, 95% CI 0.69–0.83), although the confidence intervals of all four models overlap. At its validation-selected threshold it detected 26 of the 34 AF cases (sensitivity 76.5%), at the cost of many false positives (specificity 61.8%, precision 18.2%). Optimising the preprocessing filters improved the cross-validated AUC of both methods but did not improve test performance, which is consistent with selection optimism and the small number of positive test cases. The models may therefore help flag patients at higher risk, but they are not accurate enough to serve as a clinical decision rule without better calibration, cost-based thresholds and external validation.

7 Limitations and future work

Several limitations apply to both methods. Configuration and threshold selection reused the same cross-validation predictions, potentially introducing selection optimism. Nested cross-validation could assess the complete selection procedure. Median and mode imputation also ignore uncertainty about missing values, while prediction scores whose calibration was not assessed should not be interpreted as validated individual risks. The simple imputation used still reduces the variability of the imputed inputs and ignores their relationships with other inputs. A model-based method such as k-NN or multiple imputation would be a natural improvement. Future work could therefore evaluate alternative imputation, calibration and cost-sensitive threshold selection using development data. Larger samples and independent external validation are needed, particularly given the small number of positive test cases.

7.1 Naive Bayes

Conditional independence may be unrealistic for correlated clinical predictors. Redundancy filtering does not guarantee independence within each outcome class, and overlapping evidence can affect posterior probabilities. The Gaussian assumptions for numerical predictors may also remain inadequate after transformation; extreme observations can influence estimated means and standard deviations.

Future work could assess alternative feature representations, transformations or non-Gaussian density estimates. Although the optimized configuration retained fewer predictors, it did not achieve higher test AUC. This difference cannot be attributed to feature reduction alone because the retained fitting patients, preprocessing settings and missingness indicators also differed.

7.2 k-Nearest Neighbours

Meaningful prediction depends on meaningful neighbourhoods. Equal predictor weighting can allow weakly informative variables to obscure useful similarities, while correlated predictors can give repeated weight to related information. Marginal relevance filters may also discard variables that are informative jointly. Predictor weighting or alternative feature-selection strategies could be investigated within training-set validation.

Treating ordinal predictors as nominal discards their ordering. Symmetric treatment of binary variables also gives shared absences the same contribution as shared presences, which may be unsuitable for some rare indicators. Order-aware encodings and justified asymmetric dissimilarities could improve the patient representation.

Range scaling remains sensitive to extreme fitting values, and capping contributions at one does not make it robust. Outlier flags were diagnostic and did not trigger removal or replacement. Alternative scaling could be evaluated, with its contribution to the mixed-variable distance defined explicitly.

Finally, the selected neighbourhood sizes may vary across training samples. Repeated validation could assess their stability. Prediction also requires distances to the retained reference patients; batch processing could reduce memory requirements as the dataset grows.

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Appendix

A Exploratory data analysis

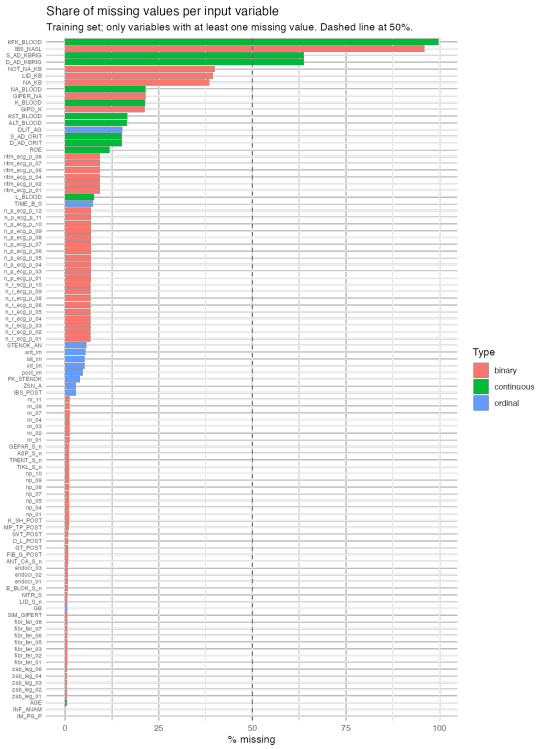


Figure A.1: Share of missing values per input variable.

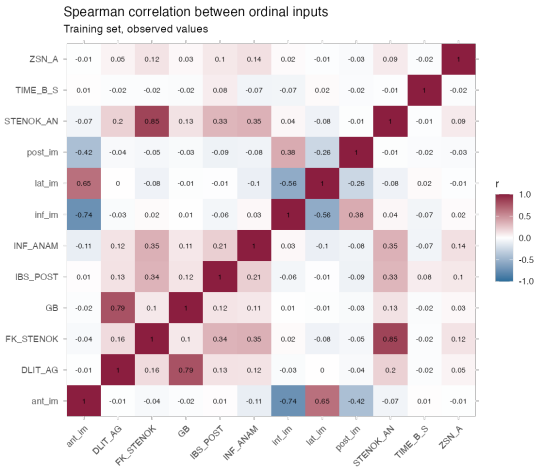


Figure A.2: Spearman rank correlation between the ordinal inputs (training set, observed values).

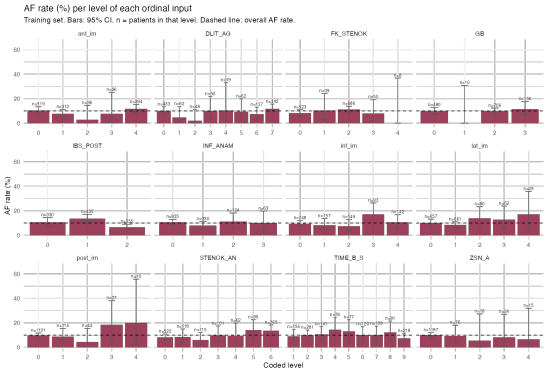


Figure A.3: AF rate per level of each ordinal input with 95% confidence interval (training set). n = patients in each level. Dashed line = overall AF rate (10%).

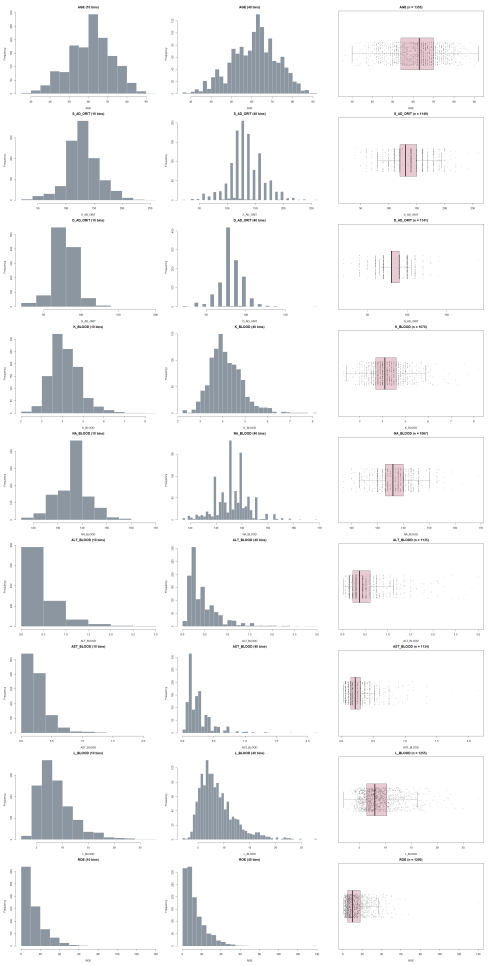


Figure A.4: Distribution of each continuous variable — two histograms (10 and 40 bins), and a boxplot of the individual observations.

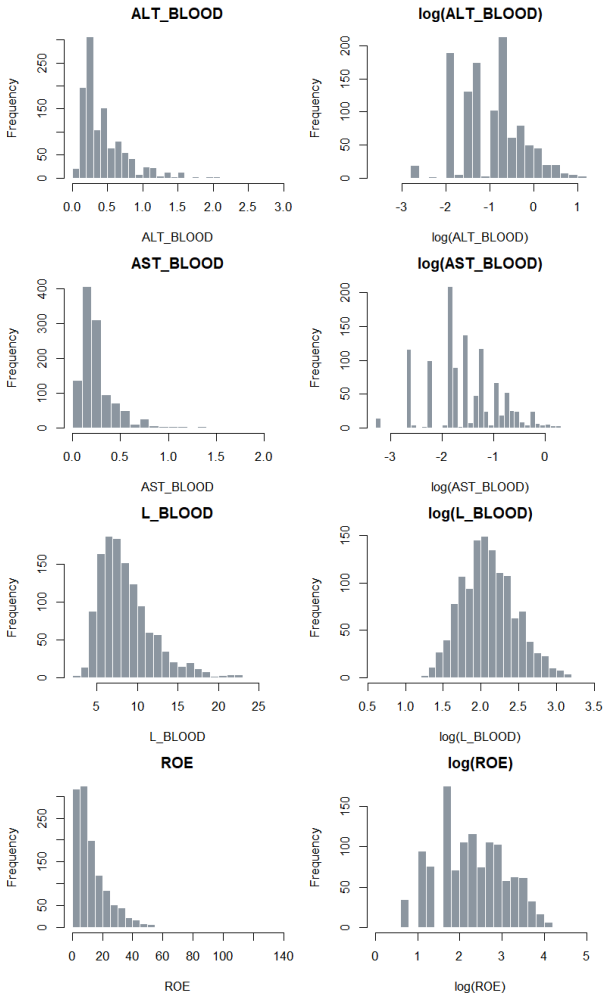


Figure A.5: Two sets of histograms showcasing the distribution of a variable and its logarithmic transform, for variables with skewness > 1.

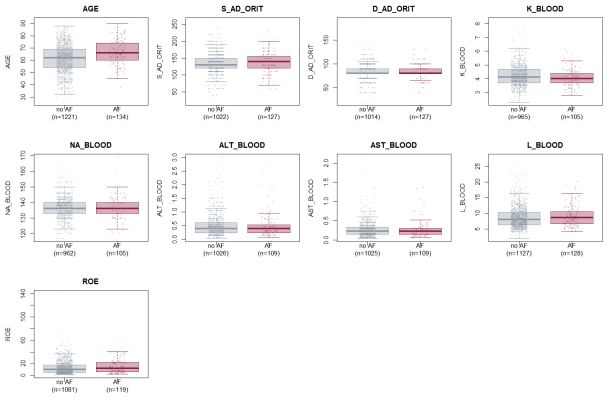


Figure A.6: Distribution of each continuous input in patients without and with AF — one boxplot per class, with each point representing a patient.

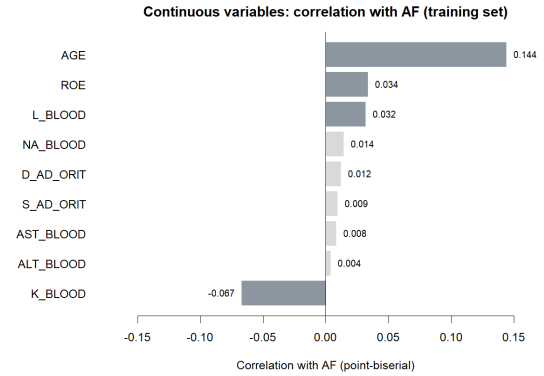


Figure A.7: Point-biserial correlation of each continuous input with AF

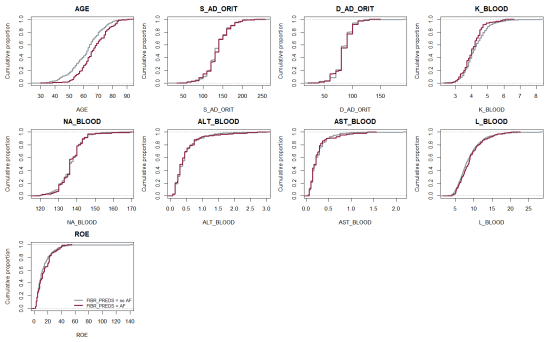


Figure A.8: Empirical cumulative distribution functions of each continuous input in patients without (gray) and with (red) AF.

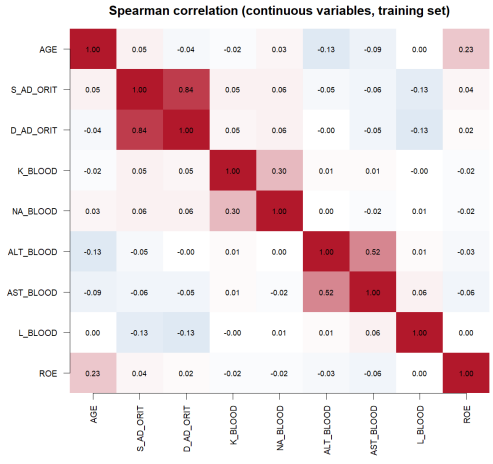


Figure A.9: Spearman correlation between the continuous inputs.

Table A.1: Binary inputs associated with AF at $p < 0.05$ (training set). AF % = AF rate among patients with value 1 or 0; Test = test used; p = unadjusted p-value; q = Benjamini–Hochberg adjusted p-value.

Variable	Description	n (value 1)	AF % if 1	AF % if 0	Test	p	FDR q
n_r_ecg_p_05	AF on admission ECG	57	43.9	8.1	chi-squared	< 0.001	< 0.001
nr_03	History of AF	32	46.9	9.2	Fisher exact	< 0.001	< 0.001
zab_leg_03	History of asthma	29	31.0	9.6	Fisher exact	0.001	0.013
nr_11	History of arrhythmia	34	29.4	9.6	Fisher exact	0.001	0.013
MP_TP_POST	AF at ICU admission	88	27.3	8.8	chi-squared	< 0.001	< 0.001
ritm_ecg_p_02	Admission ECG rhythm: AF	73	26.0	8.8	chi-squared	< 0.001	< 0.001
ritm_ecg_p_01	Admission ECG rhythm: normal sinus (60-90/min)	825	8.6	12.2	chi-squared	0.044	0.373
n_r_ecg_p_01	Premature atrial contractions on admission ECG	41	26.8	9.1	Fisher exact	0.001	0.013
SEX	Male sex	858	7.7	13.9	chi-squared	< 0.001	0.003

Table A.2: Pairs of binary or ordinal inputs with $|\rho| \geq 0.7$ (training set, Spearman rank correlation on pairwise complete observations, pairs with at least 50 patients).

Input 1	Input 2	Types	ρ	Both inputs observed (n)
FK_STENOK	STENOK_AN	ordinal / ordinal	0.85	1277
MP_TP_POST	ritm_ecg_p_02	binary / binary	0.85	1224
DLIT_AG	GB	ordinal / ordinal	0.79	1152
ritm_ecg_p_01	ritm_ecg_p_07	binary / binary	−0.78	1234
ant_im	inf_im	ordinal / ordinal	−0.74	1276
n_r_ecg_p_06	nr_04	binary / binary	0.71	1252

Table A.3: Ordinal inputs: number of levels, sparse levels (fewer than 17 training patients), range of AF rates across the other levels, Spearman rank correlation with AF (ρ), and the association test comparing the AF rate across all levels (chi-squared test; Fisher exact = Fisher's exact test, used when expected counts are below 5). Training set, observed values.

Variable	Description	Levels	Sparse	AF % range	ρ	Test	p
IBS_POST	CHD before admission	3	0	6.7–13.6	−0.070	chi-squared	0.001
inf_im	Inferior infarct (ECG)	5	0	7.4–17.2	0.019	chi-squared	0.124
STENOK_AN	Angina: how long ago	7	0	6.1–14.1	0.070	chi-squared	0.131
lat_im	Lateral infarct (ECG)	5	0	8.6–17.2	0.019	Fisher exact	0.229
post_im	Posterior infarct (ECG)	5	1	4.5–18.5	0.002	Fisher exact	0.231
ant_im	Anterior infarct (ECG)	5	0	2.8–11.7	0.009	Fisher exact	0.270
DLIT_AG	Duration of hypertension	8	0	2.1–11.7	0.019	Fisher exact	0.357
FK_STENOK	Angina functional class	5	1	8.0–11.2	0.035	Fisher exact	0.459
INF_ANAM	Number of previous infarctions	4	0	7.9–11.3	−0.020	chi-squared	0.516
TIME_B_S	Time from onset to hospital	9	0	7.3–14.5	−0.010	chi-squared	0.772
GB	Hypertension stage	4	1	9.8–11.5	0.014	Fisher exact	0.803
ZSN_A	Chronic heart failure	5	1	5.6–10.1	−0.019	Fisher exact	1.000

Table A.4: Continuous inputs by AF status — number of patients with the input recorded (n), median and interquartile range (IQR). AUC = area under the ROC curve obtained by using the input alone to separate AF from no AF (0.5 = no separation).

	n		Median		IQR		
Variable	No AF	AF	No AF	AF	No AF	AF	AUC
AGE	1221	134	62	66	15	14	0.632
K_BLOOD	965	105	4.1	4.0	1.0	0.7	0.437
ROE	1081	119	10	12	13	16	0.541
L_BLOOD	1127	128	8.0	8.6	4	3.925	0.537
AST_BLOOD	1025	109	0.22	0.22	0.18	0.15	0.481
ALT_BLOOD	1026	109	0.38	0.38	0.38	0.29	0.482
S_AD_ORIT	1022	127	130	140	30	35	0.515
D_AD_ORIT	1014	127	80	80	10	10	0.514
NA_BLOOD	962	105	136	136	7	7	0.502

B Preprocessing

Table B.1: Missingness and AF for the 49 inputs with at least 5% missing values (training set): share missing, AF rate among patients with the input missing vs. recorded, Fisher's exact test (p) and Benjamini–Hochberg adjusted p-value (q). Sorted by p. † indicates inputs with more than 50% missing values, excluded by filter 1 in the baseline configuration.

Input	Missing (%)	AF % if missing	AF % if recorded	p	q
D_AD_ORIT	16.1	4.1	11.1	< 0.001	0.04
S_AD_ORIT	15.5	4.3	11.1	0.002	0.04
LID_KB	39.4	11.9	8.7	0.064	0.40
IBS_NASL†	95.9	10.3	3.6	0.114	0.40
NA_KB	38.5	11.7	9.0	0.114	0.40
NOT_NA_KB	39.9	11.6	8.9	0.116	0.40
n_r_ecg_p_01	6.8	14.0	9.7	0.207	0.40
n_r_ecg_p_02	6.8	14.0	9.7	0.207	0.40
n_r_ecg_p_03	6.8	14.0	9.7	0.207	0.40
n_r_ecg_p_04	6.8	14.0	9.7	0.207	0.40
n_r_ecg_p_05	6.8	14.0	9.7	0.207	0.40
n_r_ecg_p_06	6.8	14.0	9.7	0.207	0.40
n_r_ecg_p_08	6.8	14.0	9.7	0.207	0.40
n_r_ecg_p_09	6.8	14.0	9.7	0.207	0.40
n_r_ecg_p_10	6.8	14.0	9.7	0.207	0.40
n_p_ecg_p_01	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_03	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_04	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_05	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_06	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_07	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_08	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_09	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_10	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_11	6.9	13.8	9.7	0.211	0.40
n_p_ecg_p_12	6.9	13.8	9.7	0.211	0.40
ALT_BLOOD	16.5	12.0	9.6	0.274	0.48
AST_BLOOD	16.6	11.9	9.6	0.276	0.48
DLIT_AG	15.2	12.1	9.6	0.313	0.53
lat_im	5.2	12.7	9.9	0.417	0.56
inf_im	5.2	12.7	9.9	0.417	0.56
STENOK_AN	5.7	12.8	9.8	0.434	0.56
ritm_ecg_p_01	9.3	11.9	9.8	0.437	0.56
ritm_ecg_p_02	9.3	11.9	9.8	0.437	0.56
ritm_ecg_p_04	9.3	11.9	9.8	0.437	0.56
ritm_ecg_p_06	9.3	11.9	9.8	0.437	0.56
ritm_ecg_p_07	9.3	11.9	9.8	0.437	0.56
ritm_ecg_p_08	9.3	11.9	9.8	0.437	0.56
L_BLOOD	7.7	7.6	10.2	0.499	0.63
ant_im	5.4	12.3	9.9	0.545	0.67
GIPO_K	21.2	10.8	9.8	0.658	0.77
K_BLOOD	21.3	10.7	9.8	0.659	0.77
S_AD_KBRIG†	63.7	10.3	9.5	0.707	0.81
GIPER_NA	21.5	10.6	9.8	0.742	0.81
NA_BLOOD	21.5	10.6	9.8	0.742	0.81
D_AD_KBRIG†	64.0	10.2	9.6	0.778	0.81
ROE	11.8	10.6	9.9	0.779	0.81
TIME_B_S	7.4	8.9	10.1	0.863	0.88
KFK_BLOOD†	99.7	10.0	0.0	1.000	1.00

Filter 4 – Relative dispersion
For each continuous variable x:

$$\mathrm{MAD}(x) = b \cdot \mathrm{median}\left(|x_i - \mathrm{median}(x)|\right), \quad b = 1.4826, \qquad D(x) = \frac{\mathrm{MAD}(x)}{\max(x) - \min(x)}.$$

The variable is removed if $D(x) \leq t_4$. The factor b makes the MAD comparable to the standard deviation under normality.

Filters 5 and 7 – Pearson correlation

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

In filter 5, y is the target coded as $0/1$, and the variable is removed if $|r_{xy}| \leq t_5$. In filter 7, x and y are two continuous variables; if $|r_{xy}| > t_7$, the one less correlated with the target is removed.

Filter 6 – Choice of test

Let X be a categorical variable with r categories and Y the target with c categories, and E_{ij} the expected counts of the contingency table:

$$\text{test} = \begin{cases} \chi^2 & \text{if } E_{ij} \geq 5 \;\; \forall\, i, j, \\ \text{Fisher's exact} & \text{if any } E_{ij} < 5 \text{ and the table is } 2 \times 2, \\ \chi^2 \text{ with Monte Carlo simulation } (B = 9999) & \text{if any } E_{ij} < 5 \text{ and the table is larger than } 2 \times 2. \end{cases}$$

The variable is removed if the p-value is $\geq t_6$.

Filters 5 and 6 – Thresholds

The thresholds of **filters 5 and 6** were chosen to test the same levels of evidence:

$$z = |r|\sqrt{n}, \qquad p = 2\,(1 - \Phi(z))\,, \tag{2}$$

where $|r|$ is the filter 5 threshold and p the filter 6 threshold. With $n \approx 1088$ (number of fitting samples in each fold), the thresholds $|r| = 0.01,\ 0.02,\ 0.05$ of filter 5 correspond to $p \approx 0.75,\ 0.50,\ 0.10$ in filter 6.

Filter 8 – Bias-corrected Cramér's V

$$\tilde{\varphi}^2 = \max\left(0,\ \frac{\chi^2}{n} - \frac{(k-1)(r-1)}{n-1}\right), \qquad \tilde{r} = r - \frac{(r-1)^2}{n-1}, \qquad \tilde{k} = k - \frac{(k-1)^2}{n-1},$$

$$\tilde{V} = \sqrt{\frac{\tilde{\varphi}^2}{\min\left(\tilde{k} - 1,\ \tilde{r} - 1\right)}},$$

where r and k are the numbers of categories of the two variables and n is the number of fitting samples. For each pair with $\tilde{V} > t_8$, the variable with the weaker association with the target is removed.

Cross-validation AUC (pooled out-of-fold) per filter, others constant

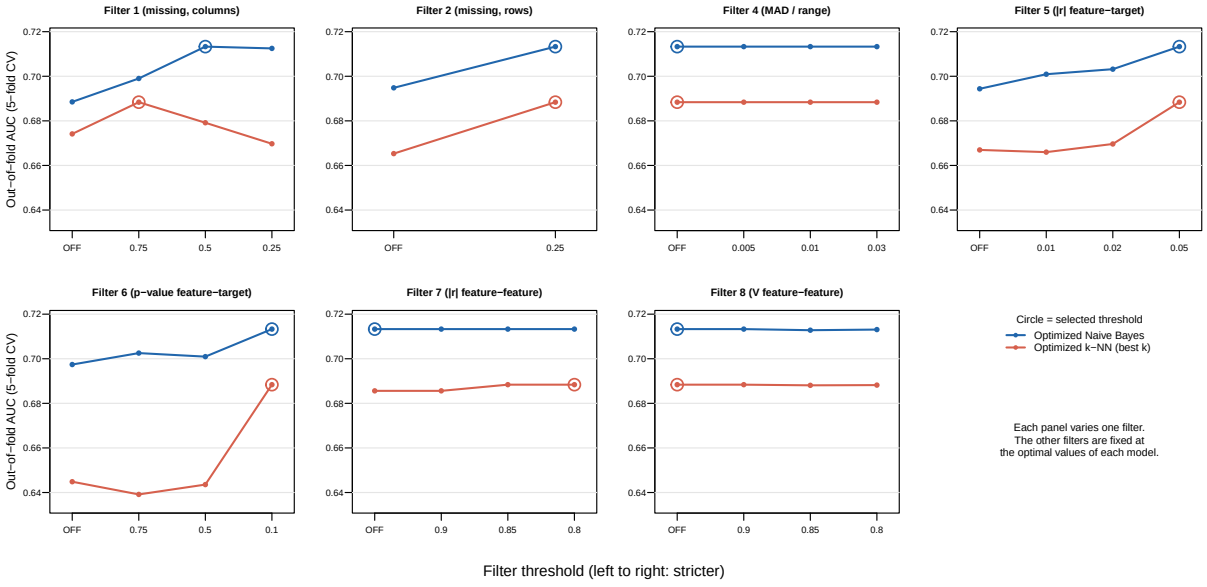


Figure B.1: Cross-validation AUC (pooled out-of-fold) when one filter is varied at a time, with the other filters fixed at the optimal values of each model. For k-NN, the best k of each configuration is used. The circle marks the selected threshold.

Outliers identified per fold

Table B.2: Fitting samples flagged by the robust Mahalanobis distance (MCD, $\alpha = 0.75$, threshold $\chi^2_{p,\,0.975}$), with the baseline configuration.

Fold	Continuous variables used	p	n fitting	Outliers	%
1	`AGE, S_AD_ORIT, K_BLOOD, L_BLOOD, ROE`	5	1086	81	7.5
2	`AGE, K_BLOOD, NA_BLOOD, L_BLOOD, ROE`	5	1087	114	10.5
3	`AGE, K_BLOOD, ALT_BLOOD, L_BLOOD, ROE`	5	1087	74	6.8
4	`AGE, S_AD_ORIT, K_BLOOD, L_BLOOD, ROE`	5	1087	76	7.0
5	`AGE, D_AD_ORIT, K_BLOOD, ALT_BLOOD, AST_BLOOD, L_BLOOD, ROE`	7	1089	97	8.9

Under multivariate normality, about 2.5% would be expected. None of these samples was removed. The continuous variables used change between folds because the filters are recomputed on the fitting data of each fold.

C Naive Bayes

C.1 Categorical probability estimation

For a categorical predictor with L_j fitting-set levels, Laplace smoothing with parameter one gives

$$\widehat{P}(X_j = \ell \mid Y = c) = \frac{n_{jc\ell} + 1}{n_c + L_j},$$

where $n_{jc\ell}$ counts fitting patients in class c with level ℓ, and n_c is the number of fitting patients in that class.

Category levels were learned exclusively from the fitting data. Categories absent from the fitting data were mapped to the corresponding fitting-set mode before prediction.

C.2 Additional results at the conventional threshold

Table C.1 reports test performance at the conventional threshold $t = 0.5$. ROC-AUC and its confidence interval are unchanged because they are calculated from prediction scores rather than a single classification threshold.

Table C.1: Naive Bayes test-set classification performance at $t = 0.5$.

Measure	Baseline	Optimized
Accuracy	0.8676	0.8618
Zero–one error	0.1324	0.1382
Sensitivity	0.2059	0.1471
Specificity	0.9412	0.9412
Balanced accuracy	0.5735	0.5441
Precision (AF class)	0.2800	0.2174
F1-score	0.2373	0.1754

C.3 Test-set confusion matrices

Figure C.1 presents the confusion matrices for both classification methods and preprocessing configurations.

For Naive Bayes at $t = 0.5$, the baseline produced 288 true negatives, 18 false positives, 27 false negatives and seven true positives. The optimized configuration produced 288 true negatives, 18 false positives, 29 false negatives and five true positives.

At the validation-selected Youden thresholds, the baseline produced 189 true negatives, 117 false positives, eight false negatives and 26 true positives. The optimized version produced 153 true negatives, 153 false positives, eight false negatives and 26 true positives.



Figure C.1: Test-set confusion matrices for Naive Bayes and k-NN with baseline and optimized preprocessing. The top row uses $t = 0.5$; the bottom row uses the Youden thresholds selected from training-set out-of-fold predictions. Within each matrix, rows represent actual classes and columns represent predicted classes.

D k-NN

D.1 Neighbourhood selection and numerical scaling

Figure D.1 shows pooled out-of-fold AUC for each candidate k. Each curve holds its preprocessing configuration fixed; it does not summarize all 8192 preprocessing combinations.

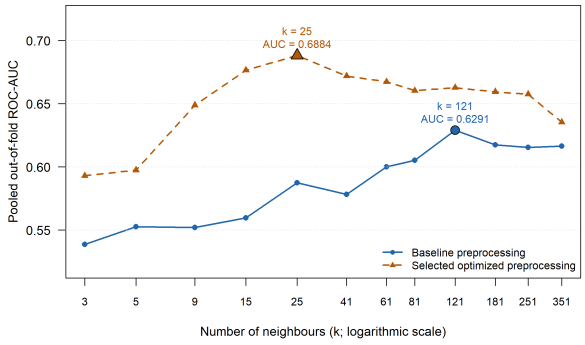


Figure D.1: Pooled five-fold out-of-fold ROC-AUC as a function of neighbourhood size, under baseline and selected optimized preprocessing. Highlighted points indicate $k = 121$ for the baseline (AUC = 0.6291) and $k = 25$ for the optimized configuration (AUC = 0.6884). The horizontal axis uses a logarithmic scale.

Numerical predictors were transformed using the retained fitting-set minimum m_j and range R_j:

$$\widetilde{x}_j = \frac{x_j - m_j}{R_j}.$$

Absolute differences on this scale equal original-scale differences divided by R_j. The same parameters were applied to validation and test observations, with each numerical distance contribution capped at one.

At the validation-selected Youden thresholds, baseline sensitivity, specificity and balanced accuracy were 0.5515, 0.6422 and 0.5968. The corresponding optimized values were 0.5294, 0.7377 and 0.6336. These are development results used during selection, rather than independent estimates of generalization.

D.2 Test ROC curves and operating points

Figure D.2 compares test ROC curves and marks the operating points corresponding to thresholds selected from training-set out-of-fold predictions.

The baseline operating point had sensitivity 0.7353 and false positive rate 0.4020. The optimized operating point had sensitivity 0.6176 and false positive rate 0.3007. No thresholds were selected using test outcomes.

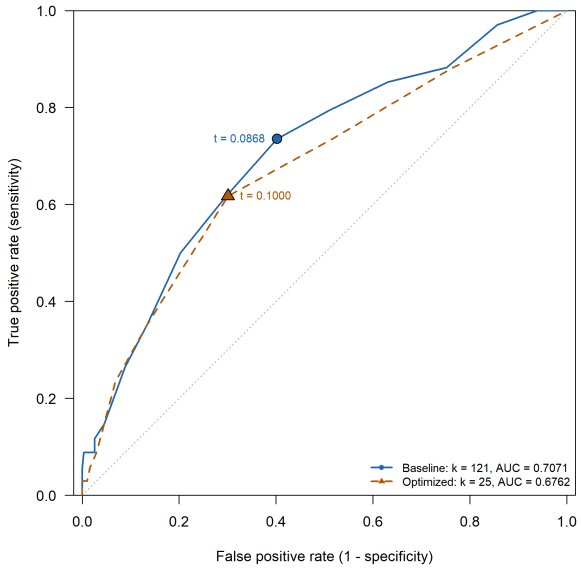


Figure D.2: Test ROC curves for baseline ($k = 121$, AUC $= 0.7071$) and optimized ($k = 25$, AUC $= 0.6762$) k-NN. Highlighted points correspond to the validation-selected Youden thresholds, applied unchanged to the test set. The grey diagonal represents chance-level discrimination.

E Results

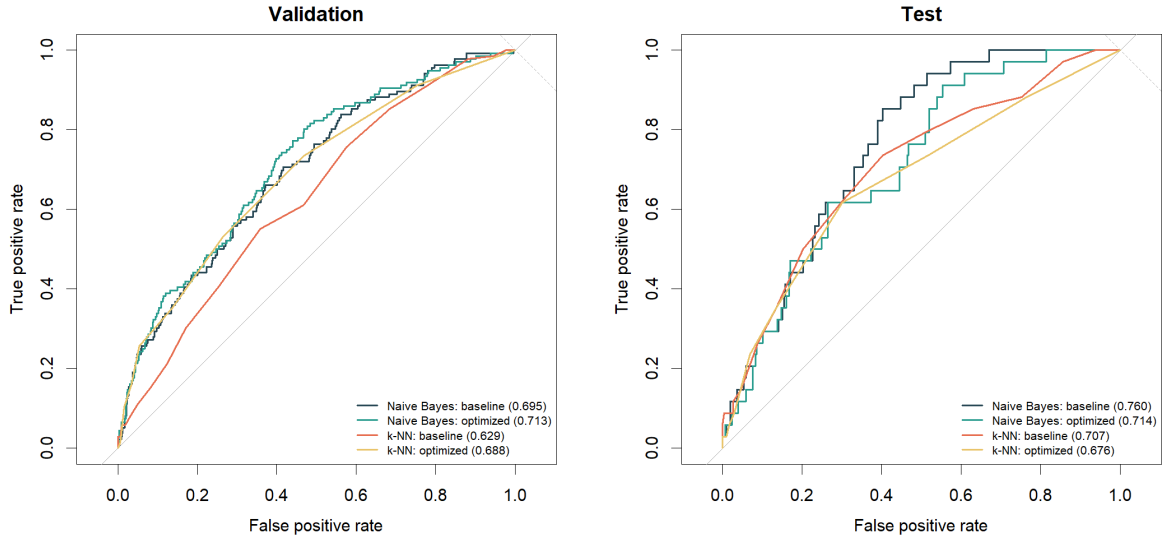


Figure E.1: ROC curves for baseline and optimized Naive Bayes and k-NN. The left panel uses pooled training-set out-of-fold predictions; the right panel uses the reserved test set. Legend values indicate the corresponding AUCs.